

STOICHIOMETRIC CALCULATIONS

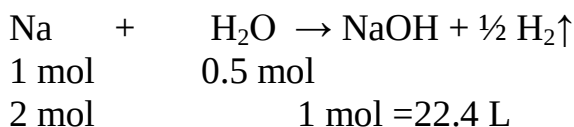
Stoichiometry is the study of the quantitative relationships among the amounts of reactants used and the amounts of products formed in a chemical reaction.



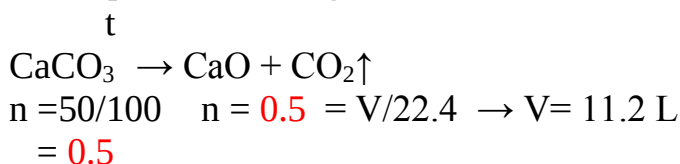
a mol of A react with b mol of B produce c mol of C

Ex: $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$ 2 moles of carbon react with 1 mole of oxygen to form 2 moles of carbon monoxide.

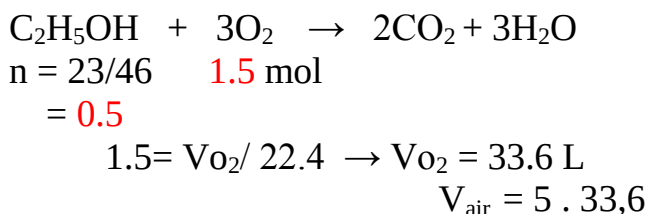
Ex : When 2 moles of sodium are put into water , what volume of hydrogen will form at STP ?



Ex: How many liters of carbon dioxide gas can be obtain from the decomposition of 50 g calciumcarbonate at STP ?



Ex: Calculate the volume of air needed for the combustion of 23 g ethanol ?
(air contains nearly 20% oxygen by volume)

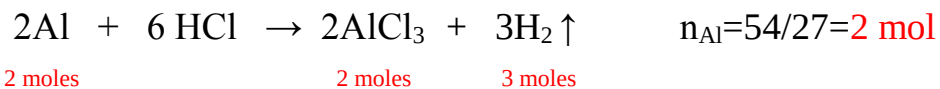


$$= 168.0$$

Ex : 54 g sample of Aluminum is reacted with hydrochloric acid ;

a) How many molecules of Aluminum chloride can form ?

b) How many liters of gas can form at STP ?



$$\begin{array}{rcl}
 \begin{array}{rcl}
 1 \text{ mol} & 6,02 \cdot 10^{23} & \\
 2 \text{ mol} & X & \\
 \hline
 x = 12,04 \cdot 10^{23} \text{ molecules of AlCl}_3 & &
 \end{array} & &
 \begin{array}{rcl}
 1 \text{ mol} & 22,4 \text{ L} & \\
 3 \text{ mol} & x & \\
 \hline
 x = 67,2 \text{ L H}_2 & &
 \end{array}
 \end{array}$$

Ex: what amount of hydrogen is produced when 5.2 mol of ammonia decomposes?

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LIMITING & EXCESS PROBLEMS

The reactant that runs out first limits the quantity of the product produced
(Limiting Reagent)

Excess Reagent a reactant that is still present after the reaction is complete.

12 desks 24 students (each can take 2 students)

both are limiting..

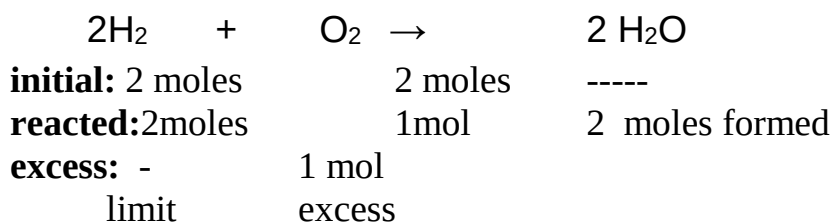
12 desk 30 student

limit excess excess part does not react

Ex:

When 2 moles of hydrogen and 2 moles of oxygen are reacted to give water,
which one becomes excess reagent?

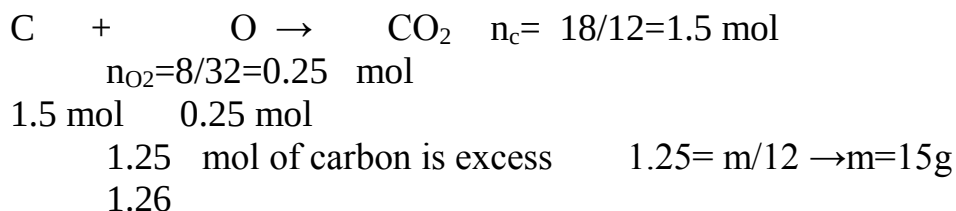
How many grams of water can be obtained at most?



$$m_{\text{H}_2\text{O}} = 2 \times 18 = 36 \text{ g}$$

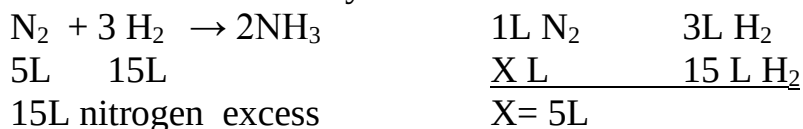
Ex:

18 g of carbon and 8 g of oxygen are reacted to form carbondioxide wwhich
element and how many grams remain behind?



Ex:

If 20 L of nitrogen and 15 L of hydrogen gases are reacted, which gas will be excess? And how many liters?



Ex:

Equal weights of sodium and chlorine are reacted to give 0.1 mol of sodium chloride. Which element remain behind? and how many grams?



initial 3.55g 3.55g

excess $m_{\text{excess Na}} = 3.55 - 2.3 = 1.25\text{g}$

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Page 332 practise 1,2,3

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PERCENTAGE YIELD

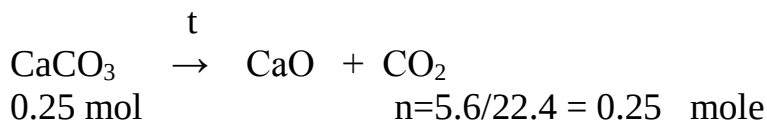
The **theoretical yield** is the amount of product that is predicted by stoichiometric calculations. However, most reactions do not produce the theoretical yield and instead produce less **actual yield**. There are several reasons why this can happen.

Percentage yield of a chemical reaction is the actual yield expressed as a percentage of the theoretical yield:

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Ex:

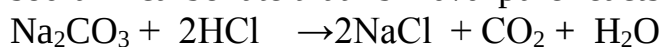
A 50 g sample of calcium carbonate is decomposed by heating and we obtain 5.6 L of carbon dioxide gas at STP . Find the **Percentage yield** of calcium carbonate sample...



$$\begin{array}{rcl} 500\text{g} & 100\% \\ 25\text{g} & x \\ \hline x & = & 50\% \end{array}$$

Ex:

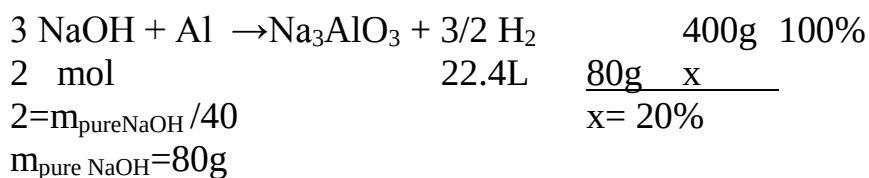
How many liters of carbon monoxide gas are released when a 53g sample of sodium carbonate that is 20% pure reacts with enough hydrochloric acid ?



$$m_{\text{pure Na}_2\text{CO}_3} = 53 \cdot 20/100 = 10.6\text{g}$$

$$0.1 = V/22.4 \rightarrow V = 2.24\text{L}$$

Ex: When 400g of sodium hydroxide solution are reacted with aluminum 22.4L of hydrogen gas at STP can be obtained . calculate the percentage yield of sodium hydroxide in the solution.



Questions;

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